

Formative-Assessment Freirean-Dialogue API for Data-Analytics Trainee-Teachers

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Abstract: *This paper presents an innovative API to help a trainee-teacher design Formative-Assessment (hereafter FA) Freirean-dialogues in a data-analytics education project. The API is emerged in context of our act to help a trainee-teacher in Germany understand what hinder the Syrian oppressed students/immigrants express freely their thoughts/critiques in group-discussions about how to initialize data-analytics education project for their self-overcoming goals. The paper shows how we interpret Freire's two works "pedagogy of the oppressed [1] and pedagogy of freedom [2]" and expose our interpretations as API's design-elements and use-value guidelines for trainee teacher's two helping functions; the first is to help him/her coordinate critical reading/analyzing circles for practicing Freirean multi-factor analysis of oppression in context of the Syrian adult oppressed learners. The second is to help him/her design and prepare FA Freirean dialogues to assess formatively (i.e., analyze critically) a limit and oppressive situation of Syrian adult learners. Moreover, it enables him/her to prepare advanced FA Freirean dialogues for analyzing critically the limit and oppressive situation and its Freirean untested feasibilities for self-overcoming. To test the API's two helping functions, test cases are suggested as next steps of this work-in-progress paper. Finally, for reflective-feedback exchange with FIE community we summarize the API's limits and our next steps to improve it.*

Keywords; *human oppression problem in Syrian context, formative assessment, data-analytics, Freirean pedagogies, Freirean untested feasibility, teacher and vocational education.*

I. INTRODUCTION : LITERATURE REVIEW AND MAIN CONTRIBUTIONS.

Assessment is usually understood to have two purposes: summative and formative. Summative assessment is used to judge students' achievements at the end of an instructional unit, whereas formative assessment aims to gather evidence about students' learning progress to influence teaching methods and priorities [3]. According to Hattie [4], using two binary notions (formative vs summative) is not helpful in education debate about assessment. To improve the debate Hattie suggests using, instead of the two notions, the single notion "Assessment-capable learner". This notion enables him to argue that both forms of assessments: summative and formative are important [4]. To support his argument, he used the metaphor suggested by Robert Stake [5] that draws not only a clear distinction line between formative and summative assessments but also highlights the importance of them both. The metaphor is: "When the cook tastes the soup, that's formative and when the guests taste the soup, that's summative". Hattie's theories of Visible Learning [6] were developed in context of primary schooling, therefore our investigations in his theories aim to test if his theories and their explaining constructs hold or do not hold in contexts of our university offers for freshman students or adult learners with different cultural and educational backgrounds. The educational philosophies of John Dewey [7], Paulo Freire [1, 2], and other advocates of progressive and critical pedagogy can be regarded, according to Noam Chomsky [8], as further

developments of Humboldt's philosophy for educational reforms started in Germany about 200 years ago. In terms of educational contexts, present-day German universities may seem far removed from the challenges and concerns that troubled Freire when he began to develop his ideas of liberatory education and pedagogy of the oppressed [1]. However, we are interested in investigating if Freire's dialogic pedagogy and its explaining constructs hold or do not hold in context of some offers provided by us for adult learners/students with immigration background because of the wars in Ukraine and in Syria. Our research contributions in [9, 10, 11] are about how to understand the pedagogy-assessment linkage and how to approach the FA implementation challenges/requirements in three different teaching or training contexts; namely context for big data-analytics training projects [10], context for a teacher of computer organization course that has about 130 freshman students [11], and context for deep-learning workshops [9]. In contrast to [9, 10, 11] in this paper we are interested in FA implementation challenges in context of Syrian oppressed adult learners. How to help trainee-teachers in Germany practice Freirean pedagogy of the oppressed to approach the human-oppression problem in context of Syrian oppressed adults is the main focus of this paper. In this paper, our main contributions are:

- We present the FA Freirean-dialogue API and its two innovative helping functions for trainee-teachers:
 1. Function for coordinating a critical reading/analyzing circle. It helps trainee-teachers in Germany practice Freirean multi-factor analysis of oppression in context of the Syrian adult immigrants/learners. It enables critical reading/analyzing of intercultural data in context of the Syrian adult immigrants' limit and oppressive situations (Section 3).
 2. Function for designing FA Freirean dialogues. It helps a data-analytics trainee-teacher design and prepare FA Freirean dialogues to assess formatively (i.e., analyze critically) a limit and oppressive situation of Syrian adult immigrants. Moreover, it helps him/her prepare advanced FA Freirean dialogues for analyzing critically the limit and oppressive situation and its Freirean untested feasibilities for self-overcoming. (Section 3).
- We present our FA dialogical practice that enables the continuous responsive-design of pedagogy and assessment acts for a self-overcoming/liberation project. This enables us to approach an ill-formulated social problem in adult education like the human oppression problem in the context of Syrian adult immigrants (Section 2).

As pre-requirement to present the innovative API and its trainee-teacher's two helping functions, in next section we present the practical problem of a German trainee-teacher in

context of summarizing the two phases of our FA dialogical practice.

II. Trainee-Teacher Practical Problem

The practical problem of a German trainee-teacher is about how to initialize a data-analytics pedagogy and assessment project for a group of Syrian oppressed students. The problem is emerged and formulated in context of the social-learning phase of our dialogical FA practice. For the sake of clarifying the problem context, the practice's two phases are presented briefly in Fig.1.

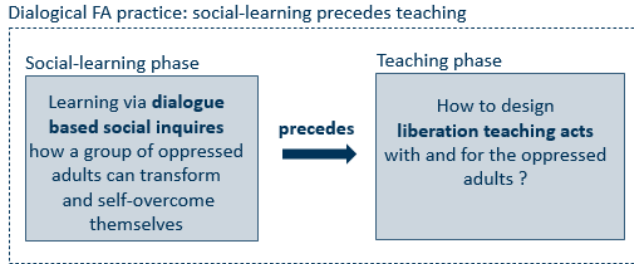


Fig. 1. Social-learning precedes teaching in our FA dialogical practice

Fig.1 shows the practice's two phases for a liberation teaching/education project. It shows that the practice's social-learning phase precedes its teaching phase. As first phase in a liberation education project, the social-learning phase uses FA dialogues as tools to enable social inquiries for investigating how a group of oppressed adults can self-overcome and transform themselves. The output of dialogue based social inquiries is the teaching phase's input in forms of liberation teaching acts with and for the oppressed adults. The teaching acts (i.e., the education project content) are not predefined and owned by the team of teachers/educators, but they are in continuous design/construction as outcomes of the dialogue based social inquiries. This enables the continuous responsive-design of the education project content as liberation teaching acts with and for the oppressed adults.

We avoid using the term “data-analytics training project” because a training project's content does not include criticizing the cultural values that domesticate and dehumanize the *lifeworld* of the Syrian adults. Criticism for deconstructing these cultural values is required as input for enabling the formation of humanist educational values. Instead of the term “training project”, we use terms like education project, teaching project or pedagogy and assessment project to emphasize that the data-analytics education project's content, under continuous responsive-design, should contain themes for studying not only the empirical dimensions in students' *social-technical reality* but also themes for studying the cultural values and the histories of oppression/domestication in this reality that hinder their self-overcoming/liberation acts.

The terms “social-technical reality” and “lifeworld” are used interchangeably in scope of the social-learning phase to enable studying und understanding a group of oppressed students and its shared lived experiences using Freire's educational philosophy. The idea that “experiences are lived and not transplanted” is a central tenet of Freire's philosophy [12]. How to investigate the lived experiences of the Syrian oppressed learners is required by the German data-analytics trainee-teacher interested in understanding what hinder the Syrian oppressed students expressing freely their critiques and thoughts in group-discussions. The group-discussions are

prepared and conducted in form of FA Freirean dialogues for enabling Freirean multi-factor analysis of oppression in the social-learning phase. For helping the trainee-teacher in preparing and designing the FA Freirean dialogues, the innovative API and its main helping functions are suggested in next section.

III. FA Freirean Dialogue API and its Main Functions

Fig. 2 shows the FA Freirean Dialogue API and its two helping functions in big picture.

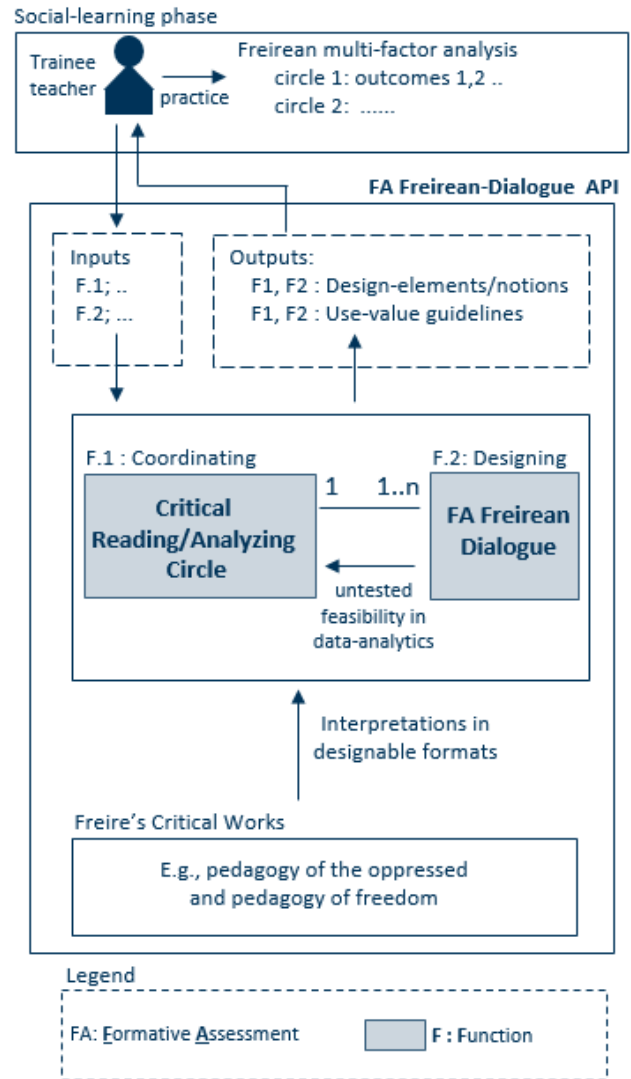


Fig. 2. FA Freirean dialogue API and its two helping functions

The Figure shows how a data-analytics trainee-teacher can practice Freirean multi-factor analysis of oppression [1] via using the two functions. The two helping functions (for brevity F.1 and F.2 as seen in Fig. 2) are our interpretations, as software designers, after reading Freire's critical works pedagogy of oppressed [1] and pedagogy of freedom [2]. As software designers we perceive the human oppression problem in context of Syrian adult learners, as a human centered design problem. The problem is how to design education as liberation act with and for a group of oppressed adults. We, as software designers, perceive or look at Freire's work “pedagogy of oppressed” as design-problem/solution to approach a target human oppression problem. The target problem in scope of this paper is the human oppression problem in context of Syrian adult immigrants/learners. In Fig.2, we summarize our understanding/interpretation of

Freire's works to approach the human oppression problem via using the API and its two helping functions to enable practicing Freirean multi-factor analysis of oppression. In brief, the first function is about enabling critical reading/analyzing circle in an intercultural Syrian-German context based on Freirean culture circle, whereas the second function is about helping a data-analytics trainee-teacher design and prepare FA Freirean dialogues to assess formatively (i.e. analyze critically) the oppressed adults' limit situations. To clarify how a trainee-teacher can practice Freirean multi-factor analysis of oppression via using the two helping functions, the two functions are described in detail in next subsections.

A. Helping Function for Coordinating a Critical Reading/Analyzing Circle.

The function is conceptualized based on reinventing Freire culture circle in context of data-analytics trainee-teacher. It is conceptualized to help German trainee-teacher coordinate a critical reading/analyzing circle for practicing Freirean multi-factor analysis of oppression in context of the Syrian adult oppressed learners. It enables critical reading/analyzing of intercultural data in context of the Syrian adult immigrants' limit and oppressive situations. Freire took education out of the classroom and created „the culture circle“, where learners used their own ways of speaking to articulate their shared understanding of how their world came to be like it was, and how to act to change their future. From being a monologue process, education became a process of dialogue in which educatees and educators engaged in mutually respectful learning. Through the culture circles process, people progressed very quickly: in Brazil illiterate adults learned to read and write in 30 hours; and in Nicaragua illiteracy was reduced from 40% to 13% in two months. Freire pointed out that the astounding results achieved by the culture circles were a consequence of offering literacy as a tool through which groups, rather than individuals, could be empowered. In culture circles, everyone makes good progress, not just a few isolated “star-students” [13]. For Freire, it is a mistake to speak of reading as solely the decoding of text. Rather, reading is a process of apprehending power and causality *in society* and one's location in it. For the scope of this paper, the term “*in society*” means in the pre-existing domesticating systems that influence and shape the decisions and the lifeworld (i.e., the lived experiences) of students and teachers. For Freire, authentic education is always a “practice of freedom” rather than an alienating inculcation of skills [14]. The benchmark that Freire used for evaluating lived experiences grew out of Christianized Marxist humanism [15]. From this position, Freire urged both students and teachers to unlearn their race, class, and gender privileges and to engage in a dialogue with those whose experiences are very different from their own. Thus, he did not uncritically affirm student or teacher experiences but provided the conceptual tools with which to critically interrogate them so as to minimize their politically domesticating influences [16]. In scope of this paper and for its main target group (i.e., the Syrian adult students), the pre-existing domesticating issues experienced by the Syrian adult learners in Syrian domesticating systems, beside the potential domesticating issues that might be emerged during communicating with German trainee-teachers are approached via using the API's coordinating function. The function's main purpose, its target user, main use-values, inputs, and outputs are summarized in table 1. Table 1 is a concrete example of the user-value

guidelines structured and exposed by the API for clarifying function 1 in Fig.2 (i.e., the coordinating function).

TABLE 1. USE-VALUE GUIDELINES FOR CLARIFYING THE COORDINATING FUNCTION

Function purpose: Coordinating critical analyzing circle to help trainee-teachers in Germany practice Freirean multi-factor analysis of oppression in context of the Syrian adult immigrants/learners.	
Reference Name: Function.1 or F.1 (see Fig.2)	
Target user: trainee-teacher as intercultural circle coordinator.	
Main Use-value: It helps data-analytics trainee teachers coordinate the critical reading/analyzing circles in intercultural Syrian-German context	
Inputs:	
1	Type: design-element Description: Input template for critical analyzing circle in intercultural Syrian-German context, • Number of Syrian adult learners in a circle • Number of German trainee-teachers
Outputs:	
1	Type: use-value guideline Description: Guideline model for clarifying Freirean multi-factor analysis of oppression. The factors are race, class, gender, culture, language, and ethnicity [1].

B. Helping Function for Designing a FA Freirean Dialogue

This function (i.e., F.2 in Fig.2) is conceptualized for helping a data-analytics trainee-teacher design and prepare FA Freirean dialogues to assess formatively (i.e., analyze critically) a limit and oppressive situation of Syrian adult immigrants. Moreover, it helps him/her prepare advanced FA Freirean dialogues for analyzing critically the limit and oppressive situation and its Freirean untested feasibilities for self-overcoming.

We agree with Freire that dialogue in pedagogy and formative assessment is very important medium to discover the reflective knowledge that distinguishes human beings. Freire justified the need for dialogue in pedagogy *epistemologically* by tracking the origin of reflective knowledge in dialogue and *ontologically* by tracking humanity in dialogue. According to Freire, communication of freely participating people, who cannot overrule themselves by authority or power, generates a special type of reflection, which only humans possess. This type of reflection is knowing about one's own knowledge. This communication (i.e., dialogue) also generates special social relations of humanity among the participants. Thus, human reflection (episteme) generates humanity, while humanity generates the reflection (i.e., one's own knowledge about his/her knowledge) [17, page 76].

Enabling the trainee-teacher to design and prepare FA Freirean dialogues will help him/her contextualize and place his/her acts as FA designer in service of supporting the Syrian adult learners to self-overcome their oppressive and limit situations.

Table 2 shows the user-value guidelines structured and exposed by the API for clarifying the designing function (i.e., F.2 in Fig.2). The table 2 shows the function's main purpose, its target user, main use-values, inputs, and outputs.

For the trainer-teacher as a designer of a FA Freirean dialogue, Fig.3 shows how we perceive the dialogue under prototyping/designing as system of descriptors and comparators. The descriptors are to capture the input data in two dimensions; dimension for limit and oppressive situation for lived expensive and dimension for untested feasibility of a situation for unlived but imaginable experience. As seen in Fig.3 the limit and oppressive situation is for a lived experience by the adult learners, whereas the untested feasible

situation is for un-lived but imaginable experience by the adult learners. For education as act that opens up the hope of transforming a limit and oppressive situation, the concept “untested feasibility of a situation” is suggested by Freire to be an object of study in context of investigating how to self-overcome and transform a limit situation.

TABLE 2. USE-VALUE GUIDELINES FOR CLARIFYING THE DESIGNING FUNCTION

Function purpose: It helps a data-analytics trainee-teacher design and prepare FA Freirean dialogues to assess formatively (i.e. analyze critically) the oppressed adults’ limit situations. Reference Name: Function.2 or F.2 (see Fig.2) Target user: Trainee-teacher as FA Freirean dialogue designer. Main Use-values: It enables a trainee teacher. design and prepare FA Freirean dialogues for analyzing critically the limit and oppressive situation and its Freirean untested feasibilities for self-overcoming.	
Inputs:	
1	Type: design-element for a Freirean analysis factor Description: setup template for a Freirean analysis factor. The factor is one of the seven analysis factors (race, class, gender, culture, language, and ethnicity)
Outputs:	
1	Type: analysis notion Description: adult learners’ limit and oppressive situation as object under analysis
2	Type: analysis notion Description: adult learners’ untested feasible situation as object under analysis
3	Type: preparation-value guideline Description: This guideline is for preparing or initializing a FA Freirean dialogue as system of generative themes [1] for description and comparison in dialogue. The initial descriptive themes generate advanced comparative themes for rational or/and Aesthetic criticism of language-uses. This guideline is explained graphically in Fig. 3.
4	Type: root-cause report-element Description: it reports the outcomes of conducting a FA Freirean dialogue for finding out what root causes behind a limit and oppressive situation.

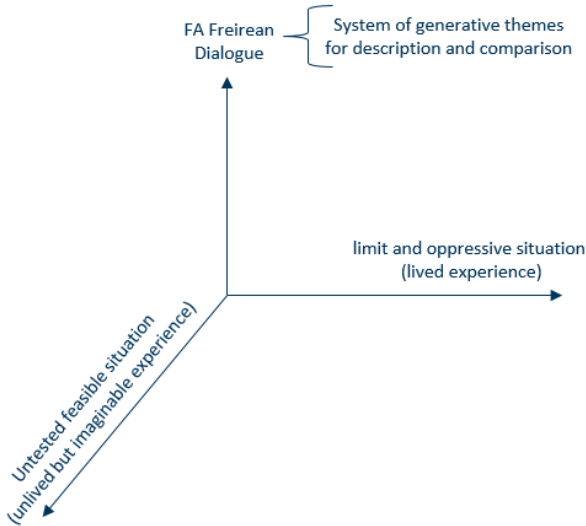


Fig. 3. Preparation-value guideline: Preparing a FA Freirean dialogue as system of generative themes for description and comparison

By using table. 2 and Fig. 3 a trainee teacher can be guided to prepare and design a FA Freirean dialogue as system of generative themes for description and comparison tasks in an adult-literacy project for programming a texts-data analytics API. The adult-literacy project is designed to help the Syrian adult-immigrants in Germany. The project focuses on investigating how to provide equitable adult-literacy support

for enabling the Syrian immigrants to read and write not only English for low-paid technical-writing jobs, but also software programs for better-paid developer jobs in Germany.

To sum up, in Fig.2, Fig.3, Table 1 and Table 2 we present design-elements, analysis-notions, use-value guidelines, exposed by the suggested API as two helping functions for trainee teachers in Germany interested in practicing Freirean multi-factor analysis of oppression in the context of Syrian adult immigrants. For testing the Freirean multi-factor analysis, in next section we present two initial test-cases.

IV. INTIAL TEST CASES FOR PRACTICING FREIREAN MULTI-FACTOR ANALYSIS

The initial test cases are about practicing Freirean multi-factor analysis to help a German trainee-teacher understand the Syrian-language uses in Syria educational context. The analysis factor “language and its uses” in Freirean multi-factor analysis is selected to coordinate two German-Syrian circles driven by the question “*how to critically analyze the Syrian-language uses experienced by Syrian adult-immigrants/students that hinder them to express freely their thoughts in group-discussions*”. The question was emerged and suggested by some Syrian adult-immigrants/students during their participation in our adult-literacy project for programming a texts-data analytics API. The test cases are initialized as two circles in tables 3 and 4. Table 3 shows the circle setups for group of Arabic Syrian immigrants/students, whereas table 4 shows the circle setups for a group of Syrian Kurdish immigrants/students.

TABLE 3. CIRCLE SETUPS FOR GROUP OF ARABIC STUDENTS

Freirean Analysis of Oppression: factor “language and it uses.” Inquiry Tools: Freirean dialogues; oral or written format Student target-group: 5 Arabic Syrian students Number of German trainee-teachers: 1	
Inputs:	
1	Prepare dialogues based on guidelines in table 2
Outputs:	
1	Collect Syrian-language uses that result in oppressive situations lived and experienced by the Arabic Syrian students

TABLE 4. CIRCLE SETUPS FOR GROUP OF KURDISH STUDENTS

Freirean Analysis of Oppression: factor “language and it uses.” Inquiry Tools: Freirean dialogues; oral or written format Student target-group: 3 Kurdish Syrian students Number of German trainee-teachers: 1	
Inputs:	
1	Prepare dialogues based on guidelines in table 2
Outputs:	
1	Collect Syrian-language uses that result in oppressive situations lived and experienced by the Kurdish Syrian students

As seen in tables 3 and 4, each circle is about finding the Syrian-language uses that result in oppressive situations lived and experienced by two groups of Syrian students. Each circle has one input and one output. The input is about preparing the FA Freirean dialogues based on the guidelines in table 2, whereas the output is about collecting Syrian-language uses that result in oppressive situations lived and experienced by the two student target-groups under study.

V. OUTCOME OF INITIAL DIALUGES

for facilitating the discussion and feedback-exchange with the FIE community, the outcome of initial dialogues is planned to be reported as structural and social facts in Syrian educational context between 1970 and 2024. To differentiate its time conditions before and during the Syrian war (started on 2011), the context is designed to have two time periods: namely, 1970-2010 Period, and 2011-2024 Period.

1970-2010 Period (Before the war): For this period the outcome of the initial dialogues shows that Syrians have multi-languages, multi-religions and multi cultures that were suppressed by the dictatorship government for 31 years before the war. It shows also that the use of Kurdish language in Syrian education organizations and systems was forbidden by systematic political strategy of the dictatorship governments. The strategy was designed to enable using only the Arabic language as oppressive education tool to suppress other languages and cultures in Syrian education systems; primary, secondary, and high education systems. For example, of an oppressive situation in this period, one master student in software engineering in Damascus university reported that he and his supervisor decided to submit in 2006 the master thesis in English. But the university rejected the English version and forced him to rewrite his thesis in Arabic, because Arabic is the only language that must be used for submitting theses in Syrian universities. Moreover, many Kurdish students reported that the exams of Arabic languages were very oppressive to them because the exams were designed in summative forms that do not enable them to get feedback to bridge their Arabic linguistic gaps.

2011-2024 Period (during the war): For this period the outcome of initial dialogues shows that like its oppression strategy before the war, the dictatorship government's strategy during the war is extremely oppressive strategy to repress its opponents in education. However, the war outcomes show that at least the Kurdish Syrians are able now to revive their language and use it in education instead of the oppressive Arabic language. Beside the previous summary of the outcome of the initial dialogues, in next weeks and before the FIE conference we plan to conduct more dialogues and share their outcomes for reflection with the FIE community.

VI. THE API LIMITS

The API first version, suggested in this paper, is limited for operating in a Syrian-German intercultural context. It enables intercultural investigation using Freirean circles and dialogues to help the Syrian immigrants integrate in Germany. It also helps the host German education communities to be more culturally aware about the human oppression problem in Syrian immigrant context. Moreover, in the adult education literature our paper helps to introduce this problem as an ill-formulated oppression problem in adult immigrant context.

VII. CONCLUSION AND FUTURE WORKS

In Fig.2, Fig.3, Table 1, and Table 2 we have presented design-elements, analysis-notions, use-value guidelines, exposed by the suggested API as two helping functions for trainee teachers in Germany interested in practicing Freirean multi-factor analysis of oppression in context of Syrian adult immigrants/learners. The FA Freirean-dialogue API's two helping functions are:

- The first function is to help a data-analytics trainee-teacher coordinate Freirean intercultural circles for enabling critical reading/analyzing of intercultural data in context of the Syrian adult immigrants' limit and oppressive situations.
- The second function is to help a data-analytics trainee-teacher design and prepare FA Freirean dialogues to assess formatively (i.e., analyze critically) a limit and oppressive situation of Syrian adult immigrants. Moreover, it helps him/her prepare advanced FA Freirean dialogues for analyzing critically the limit and

oppressive situation and its Freirean untested feasibilities for self-overcoming.

Our future works are to plan and conduct test-cases not only for the analysis factor "language" as presented in Tables 3 and 4, but also for the other analysis factors (race, class, gender, culture, and ethnicity) for improving the quality of the Freirean multi-factor analysis. We plan also to integrate this work with our previous work [18] that uses flow theory and Freirean pedagogy in workshops settings. Integrating and reporting this work for German research communities in vocational education like in [19] is also planned.

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